

Assignment - Data Aggregation Activity (10%)

Due: Module 5

Purpose:

The purpose of this assignment is to aggregate data in a format for analysis and extract those data for further problem solving.

Assignment Objective:

For the business problem you selected, perform the required steps to aggregate data for visualization using a series of summarizations, including: Count and Average.

Instructions:

This activity relates to the Video Games Sales Data:

Click on the link below to download your data:

- [Video Games Sales - Data](#)

In this assignment you will aggregate the data for extracting insights (you might need to reload the data in HDInsight and Hive). Do the following steps:

- 1) Create 2 tables. The first table shows the average critic score for Sports games. Store this information in a column called "sports_critic_score". The second table shows the average critic score for Shooter games. Store this information in a column called "shooter_critic_score".
- 2) Create 3 statistics tables (average, min, max) for the global_sales for: all games, Sports games, Shooter games. For each table, label the columns as: "average_global_sales", "min_global_sales", and "max_global_sales".
- 3) Create 2 tables containing the global_sales and the count of games with that global_sales for: Sports games and Shooter games. For each table, label the columns as: "global_sales" and "count".

This assignment relates to the following Course Learning Requirements:

CLR 2: Illustrate how current tools are used to address these problems, including the strength and weaknesses of each tool

CLR 4: Extract and load data to support problem solving and decision-making

Grading for This Assignment

Please take a screen shot of each step, showing the result. Also provide the query you used.

Steps required	Grade Earned
1) Create two tables (one for sports games and one for shooter games) displaying the average critic score	50%
2) Create three statistics tables for the global sales; one for all games, one for sports games and one for shooter games	80%
3) Create two tables (one for sports games and one for shooter games) for global sales and the number of games matching that global sales amount	100%

****These tables were created from the point where the last assignment ended, the transformations from Assignment #1 were performed first and then the tables below were created****

1)

Query and Table 1: Sports Games displaying Average Critic Score

MentalHealthSurvey

VideoGameSales *

+

DATABASE

Select or search database/schema

default

```

1 SELECT AVG(Critic_Score) AS Sports_Critic_Score
2 FROM sales_genre
3 WHERE Genre LIKE "%Sports%";

```

✓ Execute

Save As

Insert UDF

Visual Explain

RESULTS

LOG

VISUAL EXPLAIN

TEZ UI

Filter columns

×

sports_critic_score

71.96311818943839

Table 2: Shooter Games displaying Average Critic Score

MentalHealthSurvey ✕
VideoGameSales * ✕
+

DATABASE
Select or search database/schema

✕ default

```

1 SELECT AVG(Critic_Score) AS Shooter_Critic_Score
2 FROM sales_genre
3 WHERE Genre LIKE "%Shooter%";

```

✓ Execute

Save As

Insert UDF ▾

Visual Explain

RESULTS

LOG

Visual Explain

TEZ UI

Filter columns ✕

shooter_critic_score
70.16932907348243

2)

Table 1: Global Sales - ALL

MentalHealthSurvey ✕
VideoGameSales * ✕
+

DATABASE
Select or search database/schema

✕ default

```

1 SELECT AVG(Global_Sales) AS Average_Global_Sales, MIN(Global_Sales)
2 AS Min_Global_Sales, MAX(Global_Sales) AS Max_Global_Sales
3 FROM sales_genre;

```

✓ Execute

Save As

Insert UDF ▾

Visual Explain

RESULTS

LOG

Visual Explain

TEZ UI

Filter columns ✕

average_global_sales	min_global_sales	max_global_sales
0.61339252682204957455	0.00	83.00

Table 2: Global Sales - Sports Games

MentalHealthSurvey ×	VideoGameSales * ×	+
DATABASE Select or search database/schema		
× default		
<pre> 1 SELECT AVG(Global_Sales) AS Average_Global_Sales, MIN(Global_Sales) 2 AS Min_Global_Sales, MAX(Global_Sales) AS Max_Global_Sales 3 FROM sales_genre 4 WHERE Genre LIKE "%Sports%"; 5 </pre>		
✓ Execute	Save As	Insert UDF ▾ Visual Explain
RESULTS	LOG	VISUAL EXPLAIN TEZ UI
Filter columns ×		
average_global_sales	min_global_sales	max_global_sales
0.69069572506286672255	0.00	83.00

Table 3: Global Sales - Shooter Games

MentalHealthSurvey ×	VideoGameSales * ×	+
DATABASE Select or search database/schema		
× default		
<pre> 1 SELECT AVG(Global_Sales) AS Average_Global_Sales, MIN(Global_Sales) 2 AS Min_Global_Sales, MAX(Global_Sales) AS Max_Global_Sales 3 FROM sales_genre 4 WHERE Genre LIKE "%Shooter%"; 5 </pre>		
✓ Execute	Save As	Insert UDF ▾ Visual Explain
RESULTS	LOG	VISUAL EXPLAIN TEZ UI
Filter columns ×		
average_global_sales	min_global_sales	max_global_sales
0.81789137380191693291	0.00	15.00

3)

Query Run:

DATABASE
 Select or search
 database/schema

×  default

```

1 SELECT Global_Sales, COUNT(Global_Sales) AS Count
2 FROM sales_genre
3 WHERE Genre LIKE "%Sports%"
4 GROUP BY Global_Sales
5 ORDER BY Global_Sales;
6

```

Table 1: Sports Games – Global Sales and the Number of Games Matching the Global Sales Amount

global_sales	count
0.00	757
1.00	306
2.00	69
3.00	30
4.00	16
5.00	6
6.00	3
8.00	1
9.00	1
22.00	1
23.00	1
33.00	1
83.00	1

Query Run:

MentalHealthSurvey ×

VideoGameSales * ×

+

DATABASE

Select or search database/schema

×
 default

```

1 SELECT Global_Sales, COUNT(Global_Sales) AS count
2 FROM sales_genre
3 WHERE Genre LIKE "%Shooter%"
4 GROUP BY Global_Sales
5 ORDER BY Global_Sales;
6

```

Table 2: Shooter Games – Global Sales and the Number of Games Matching the Global Sales Amount

global_sales	count
0.00	584
1.00	221
2.00	56
3.00	30
4.00	16
5.00	5
6.00	5
7.00	6
8.00	2
9.00	2
10.00	3
11.00	1
12.00	1
13.00	3
14.00	2
15.00	2